Homework 12.10.26 – Exercises 9 and 10

Gor Melkumyan

Exercise 9

Given $\mu I \preceq \nabla^2 f \preceq LI$, $0 < \mu \leq L$, and $T_\gamma(x) := x - \gamma \nabla f(x)$.

1)

$$T_\gamma(x) - T_\gamma(y) = \int_0^1 \nabla T_\gamma\big(y + t(x-y)\big)(x-y)\, dt \;\Rightarrow$$

$$\begin{aligned}\|T_\gamma(x) - T_\gamma(y)\| &= \left\| \int_0^1 \nabla T_\gamma\big(y + t(x-y)\big)(x-y)\, dt \right\| \leq \int_0^1 \left\|\nabla T_\gamma\big(y + t(x-y)\big)\right\| \|x-y\|\, dt \\ &\leq \int_0^1 \sup_{z\in\mathbb{R}^d} \|\nabla T_\gamma(z)\|\, dt \cdot \|x-y\| = \sup_{z\in\mathbb{R}^d} \left\|I - \gamma\nabla^2 f(z)\right\| \|x-y\|.\end{aligned}$$

Since $\mu I \preceq \nabla^2 f \preceq LI$, the eigenvalues λ_i of the symmetric matrix $\nabla^2 f(z)$ lie in $[\mu, L]$, so

$$\left\|I - \gamma\nabla^2 f(z)\right\| = \max_i |1 - \gamma\lambda_i| \leq \max\{|1-\gamma L|, |1-\mu\gamma|\}$$

($|1-\gamma\lambda|$ is convex in λ, so its maximum on $[\mu, L]$ is at an endpoint) $\Rightarrow$

$$\|T_\gamma(x) - T_\gamma(y)\| \leq \|x-y\| \cdot \max\{|1-\gamma L|, |1-\mu\gamma|\}.$$

2)

If $\gamma = \frac{2}{\mu+L} \Rightarrow |1-\gamma L| = |1-\mu\gamma| = \frac{L-\mu}{L+\mu}$.

If $\gamma > \frac{2}{\mu+L} \Rightarrow L\gamma > \frac{2L}{\mu+L} \Rightarrow 1 - L\gamma < 1 - \frac{2L}{\mu+L} = \frac{\mu-L}{\mu+L} \leq 0 \Rightarrow |1-L\gamma| > \frac{L-\mu}{L+\mu} \Rightarrow \max\{|1-\gamma L|, |1-\mu\gamma|\} > \frac{L-\mu}{L+\mu}$.

If $\gamma < \frac{2}{\mu+L} \Rightarrow \mu\gamma < \frac{2\mu}{\mu+L} \Rightarrow 1 - \mu\gamma > 1 - \frac{2\mu}{\mu+L} = \frac{L-\mu}{L+\mu} \geq 0 \Rightarrow |1-\mu\gamma| > \frac{L-\mu}{L+\mu} \Rightarrow \max\{|1-\gamma L|, |1-\mu\gamma|\} > \frac{L-\mu}{L+\mu}$.

$\Rightarrow \gamma = \frac{2}{\mu+L}$ is optimal.

3)

Since $\nabla f(x^\star) = 0$, we have $T_\gamma(x^\star) = x^\star$, so with $\gamma = \frac{2}{\mu+L}$

$$\left\|x^{k+1} - x^\star\right\| = \left\|T_\gamma(x^k) - T_\gamma(x^\star)\right\| \leq \frac{L-\mu}{L+\mu}\left\|x^k - x^\star\right\| \;\Rightarrow\; \left\|x^k - x^\star\right\| \leq \left\|x^0 - x^\star\right\| \cdot \left(\frac{L-\mu}{L+\mu}\right)^k.$$

In Theorem 15, $\left\|x^k - x^\star\right\|^2 \leq (1-\gamma\mu)^k \left\|x^0 - x^\star\right\|^2$, where $\gamma \leq \frac{1}{L}$; the best choice is $\gamma = \frac{1}{L}$

$$\Rightarrow \left\|x^k - x^\star\right\|^2 \leq \left(1 - \frac{\mu}{L}\right)^k \left\|x^0 - x^\star\right\|^2 = \left(\frac{L-\mu}{L}\right)^k \left\|x^0 - x^\star\right\|^2.$$

Squaring our bound, $\left\|x^k - x^\star\right\|^2 \leq \left(\frac{L-\mu}{L+\mu}\right)^{2k} \left\|x^0 - x^\star\right\|^2$, and

$$\frac{L-\mu}{L} : \left(\frac{L-\mu}{L+\mu}\right)^2 = \frac{(L+\mu)^2}{L(L-\mu)} > 1 \quad (\mu < L),$$

so the rate with $\gamma = \frac{2}{\mu+L}$ is better, by the factor $\frac{(L+\mu)^2}{L(L-\mu)}$ per iteration.

Exercise 10

If f is L-smooth and μ-strongly convex, then by (18) and Exercise 8

$$\frac{1}{L}\left\|\nabla f(x^k) - \nabla f(x^\star)\right\|^2 \leq 2D_f(x^k, x^\star) \leq \frac{1}{\mu}\left\|\nabla f(x^k) - \nabla f(x^\star)\right\|^2.$$

Dividing by $\left\|\nabla f(x^k) - \nabla f(x^\star)\right\|^2$, (i) and (ii) are true: $\frac{1}{L} \leq \gamma_k \leq \frac{1}{\mu}$.

For (iii): by (i), $1 - \gamma_k L \leq 0$, so the last term of Lemma 14 cannot be dropped. Instead we stop the proof of Lemma 14 before using (25), with $r^k = x^k - x^\star$:

$$\left\|r^{k+1}\right\|^2 \leq (1 - \gamma_k\mu)\left\|r^k\right\|^2 - 2\gamma_k D_f(x^k, x^\star) + \gamma_k^2\left\|\nabla f(x^k) - \nabla f(x^\star)\right\|^2.$$

By the definition of γ_k, $\gamma_k^2\left\|\nabla f(x^k) - \nabla f(x^\star)\right\|^2 = 2\gamma_k D_f(x^k, x^\star) \Rightarrow$

$$\left\|x^{k+1} - x^\star\right\|^2 \leq (1 - \gamma_k\mu)\left\|x^k - x^\star\right\|^2.$$